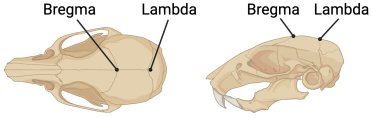
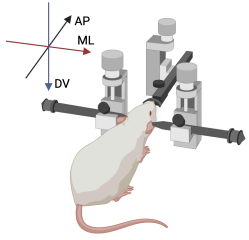
A

```
sub-mouse01/ecephys/
├── sub-mouse01_probes.tsv                                  ← stereotaxic
├── sub-mouse01_electrodes.tsv                              ← probe-relative
├── sub-mouse01_space-AllenCCFv3_electrodes.tsv             ← atlas
└── sub-mouse01_space-AllenCCFv3_coordsystem.json           ← atlas
```

B *Stereotaxic coordinate system*



Bregma & Lambda reference points



AP / ML / DV axes

C *Probe-relative coordinate representation*

***_electrodes.tsv**

name	probe_name	x	y	z	shank_id	location
e001	probe01	0	0	0	0	MOp
e002	probe01	0	0	25	0	MOp
e033	probe02	0	0	0	0	CA1

D *Atlas coordinates (with space entity)*

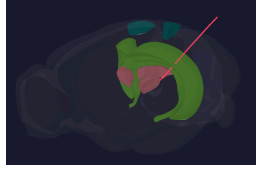
***_space-AllenCCFv3_electrodes.tsv**

name	probe_name	x	y	z
e001	probe01	5700	4200	3800
e002	probe01	5700	4200	3775
e033	probe02	7800	3500	2800

space-AllenCCFv3_coordsystem.json

```
{ "MicroephysCoordinateSystem": "AllenCCFv3",
  "MicroephysCoordinateUnits": "um",
  "MicroephysCoordinateSystemDescription":
    "Origin at anterior commissure…" }
```

x, y, z = position in brain atlas (μm)



***_probes.tsv** *Surgical coordinates per probe*

probe_name	AP	ML	DV	AP_angle	ML_angle	rotation_angle	anatomical_reference_point	associated_brain_region	reference_atlas
probe01	-2.5	1.5	-4.0	15	0	0	Bregma	Primary Motor Cortex	AllenCCFv3
probe02	-1.8	2.0	-3.5	10	5	0	Bregma	Hippocampus CA1	AllenCCFv3